David Joy

+1 678-873-0295 | davidjoydev@gmail.com | www.linkedin.com/in/david-joy-me | github.com/davidjoyme

Education

University of Georgia(UGA) — **Athens,USA**
Major: Artificial Intelligence -Thesis Track, GPA: 3.82/4.00 — ***Jan 2024 – Aug 2026***

- Relevant Coursework: Artificial Intelligence, Data Mining, Privacy-Preserving Data Analysis,Computer Vision

National Institute of Technology Calicut (NITC) — **Kozhikode, India**
Major: Electronics and Communication Engineering, CGPA: 7.64/10 — ***Jul 2019 – May 2023***

- Relevant Coursework: Probability & Statistics, C++ Programming, Python Programming, Data Structures & Algorithms, Digital Signal Processing

Experience

MODEL Lab - University of Georgia — **Mar 2024 – Present**
Research Assistant — *Athens,Georgia*

- Research Assistant at the **Manufacturing Optimization Design Engineering-Education Lab(MODEL) under Dr. Beshoy Morkos** in developing computer vision and LLM AI models to accelerate smart manufacturing.
- Set up **end-to-end pipelines** to develop **intelligent machines** in the manufacturing sector for **Middleby** industrial equipments that utilizes computer vision and image segmentation to **increase yield by 50%**.
- Engineering team **member of the Georgia - AI for Manufacturing (GA-AIM) project** funded by the U.S Economic Development Administration (EDA) in creating scalable local ML applications for **100+ manufacturers**

Indian Institute of Science — **May 2023 – Dec 2023**
Machine Learning Researcher — *Bengaluru,Karnataka*

- Machine Learning Researcher in the **web applications team** of the **Design and Innovation Centre** at IISc. Bangalore.
- Incorporated **automated video segmenting** using **computer vision** on video screencast recordings of collaborative videos
- Implemented **Siamese Neural Networks** for **transfer learning** for video action recognition tasks that replaced manual labeling.

Projects

Retail Price Optimization & Causal Impact Engine (Rossmann Stores) | *Python,SQLite* — **May 2025 - Aug 2025**

- Built a retail pricing intelligence system using **XGBoost forecasting + EconML** causal inference on the 1M+ row Rossmann dataset, estimating promo-driven uplift (ATE/CATE) and deriving store-specific price elasticity.
- Engineered **SQL-based feature pipelines** (SQLite + Pandas) and developed a **real-time pricing optimizer that models revenue vs. price curves**, identifies optimal price points, and supports future event simulations.

Smart Machine Operator AI Agent | *Python,LLM,LangChain,Flask* — **Dec 2024 - Jan 2025**

- Programmed a **smart voice assistant AI agent** capable of **retrieving information** from lengthy instruction manuals by citing sources, and capable of **controlling IoT Machines** and their **digital twins** built on Unity 3D.
- Experimented with **RAG and LLM Fine-tuning** in local **LLaMA and Mistral models** with SQLLite caching accompanied by tool calling for **controlling external APIs**.
- Implemented **full-stack voice interactive pipeline** using Whisper(Speech to Text),Flask, Piper(Text to Speech) and ChromaDB.**Automated data visualization charts** from command prompts based on user requirements.

3-D CAD Generative Model | *PyTorch,Python* — **Sep 2025 - Jan 2026**

- Developed a self-supervised **3-D Generative machine learning model** for manufacturing-based components using latent space flow matching on CAD model datasets.
- Utilized Mamba State Space models to achieve **State-of-the-art metrics** for 3D CAD solid generation from the AutoCAD ABC dataset.

Publications

- A Comparative Study Between Humans and GPT for Tradespace Analysis - ASME IDETC 2025

Technical Skills

Languages and Skills: Python, C++, SQL, Machine Learning, R, Computer Vision, Multi-Agent Systems, MLOps
Developer Tools: Linux bash, Git,Docker, ONNX, MLflow, Airflow, DVC, AWS, Azure, Databricks, SnowFlake.
Technologies/Frameworks: PyTorch, Tensorflow, LangChain, LangGraph, Flask, OpenCV